Sequences — problem 9.50

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Given are the positive integers $a_0, \ldots, a_{100}$ such that $a_1 > a_0$, $a_2 = 3a_1 - 2a_0$, $a_3 = 3a_2 - 2a_1, \ldots, a_{100} = 3a_{99} - 2a_{98}$. Prove that $a_{100} > 2^{99}$.